

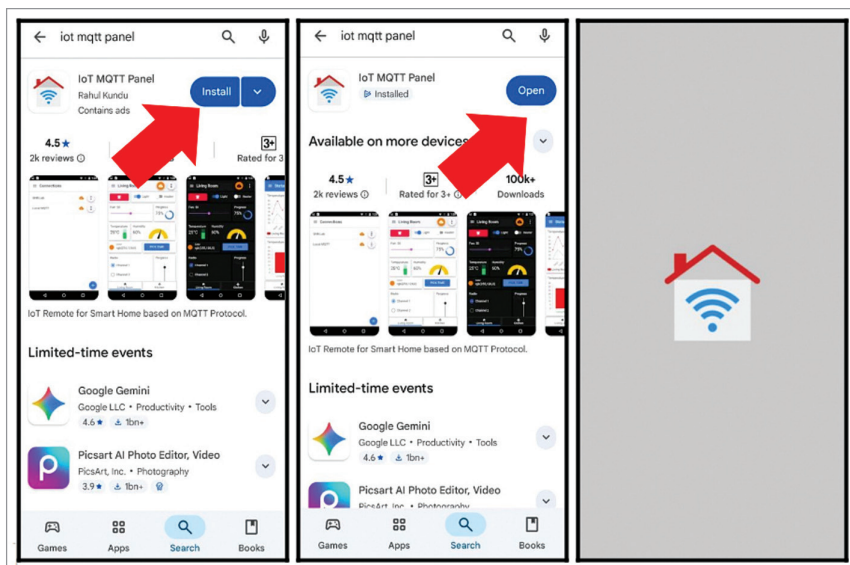


Fig. 8: Installing the IoT MQTT panel app

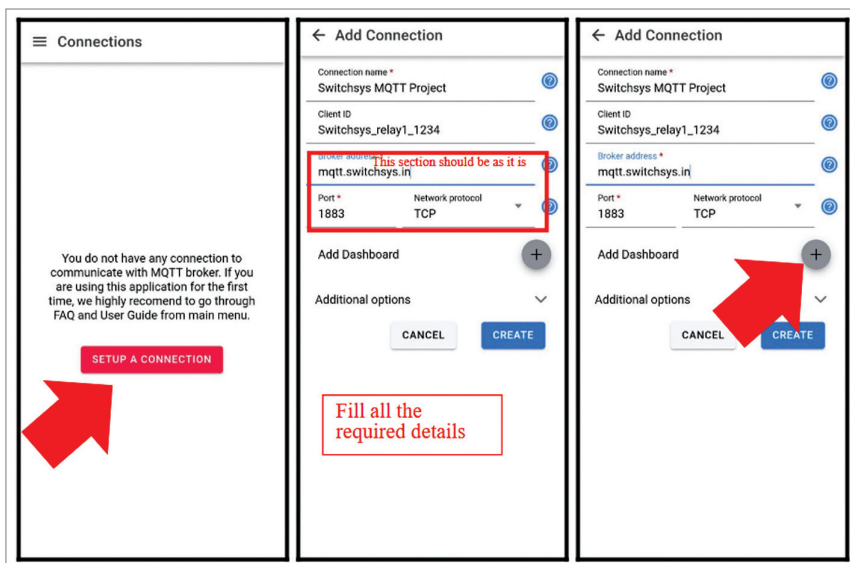


Fig. 9: MQTT configuration

EFY note. This system uses the ESP-01 module, and the GPIO pins are selected accordingly. Other ESP8266-based boards, such as NodeMCU and WeMos D1 Mini, can also be used with this system. In that case, only the ESP-01 module needs to be replaced, and the GPIO pin connections should be modified accordingly.

Software

The first step is to configure the MQTT broker used for communication between the IoT MQTT mobile application and the ESP-01 module.

In this system, the Switchsys MQTT Broker, a free public MQTT broker designed for learning, testing, and IoT prototyping, is used.

The broker supports standard MQTT communication over TCP using port 1883 and does not require authentication, making it suitable for learning, testing, and evaluating IoT systems without the need for dedicated server infrastructure. The broker details used in this system are as follows:

Broker URL: `mqtt.switchsys.in`
and Port: 1883 using non-TLS MQTT communication.

Next, the firmware must be prepared for the ESP module. In this system, the ESP-01 module is used. The ESP-01 is a compact, low-cost Wi-Fi module based on the ESP8266 System-on-Chip (SoC), which includes an integrated TCP/IP protocol stack. Due to its small size, built-in Wi-Fi capability, and ease of programming, it is widely used in standalone IoT applications. The module comes with 1MB flash memory and provides two usable GPIO pins, namely GPIO0 and GPIO2. The ESP-01 can be programmed using the Arduino IDE with the help of an ESP-01 USB Programmer Adaptor.

In this system, the ESP-01 receives commands from the IoT MQTT application through the Switchsys MQTT broker and controls the relay connected via the ULN2003A relay driver circuit. As shown in the circuit diagram in Fig. 2, GPIO0 is used as the relay control pin. The source code can be downloaded by scanning the accompanying QR code. After downloading, open the file named `IoT_Smart_Relay_Switchsys_MQTT_IoT_MQTT_Panel.ino` in the Arduino IDE. In the code, the relay control pin, Wi-Fi credentials, and MQTT settings must be configured according to the network and application requirements.

The relay control pin configuration used in this system is shown in Fig. 3.

Note. This system is designed using the ESP-01 module, and the GPIO pin selection is based accordingly. Other ESP8266-based boards, such as NodeMCU or WeMos D1 Mini, can also be used. In that case, the GPIO pin definitions must be modified according to the selected development board.

The Wi-Fi credentials section of the code must also be updated according to the local Wi-Fi network being used. The relevant code section for Wi-Fi configuration is shown in Fig. 4.